



Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune



Presents



Where Ideas Become Reality

Team Name : **TechNova**

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Selected Domain : **Agri-Tech**

PROBLEM STATEMENT

❖ Problem Statement

- Poor storage infrastructure causes major post-harvest losses.
- Lack of temperature and humidity monitoring accelerates spoilage.
- No real-time inventory tracking leads to mismanagement.
- Inefficient supply chain coordination delays distribution.
- Results in food wastage, farmer losses, and price inflation.
- Delays in distribution increase waste.



App Logo



Intelligent Monitoring and Distribution Planning to Minimise Post-Harvest Losses

• Proposed Solution

- Real-time monitoring of temperature and humidity using IoT sensors
- Centralized dashboard for live inventory and batch tracking
- AI-based spoilage prediction to identify high-risk produce
- Automated alert system for preventive action
- Intelligent distribution planning to optimize stock movement and reduce waste

Diagnoses the Problem	Innovation & Uniqueness
Continuous Environmental Monitoring	 • IoT + AI for S
Spoilage Alerts	 • Smart Redist
Real-time Inventory Visibility	 • Data-Driven
Optimized Distribution	 • Scalable for I



- AI-powered system for real-time monitoring and inventory tracking
- Predicts spoilage early and optimizes distribution planning
- Reduces food waste, farmer losses, and supply chain inefficiencies
- Sends automated alerts for temperature and humidity changes
- Improves transparency and coordination across warehouses

Technical Approach (Hackathon Summary)

Tech Stack	2. The Core Process	3. Impact
IOT/Hardware ESP32, DHT11 RFID/QR	  	
Backend	1. Harvest Entry	 Zero Visibility
Backend Node.js/Python	2. Smart Monitoring (AI alerts)	 Reduced Spoilage (30-40% reduction)
Frontend	3. Dynamic Inventory (FEFO)	
React Native App Regression/Genetic Algos	4. Optimized Exit (Routes)	 Faster Turnover

FEASIBILITY AND VIABILITY



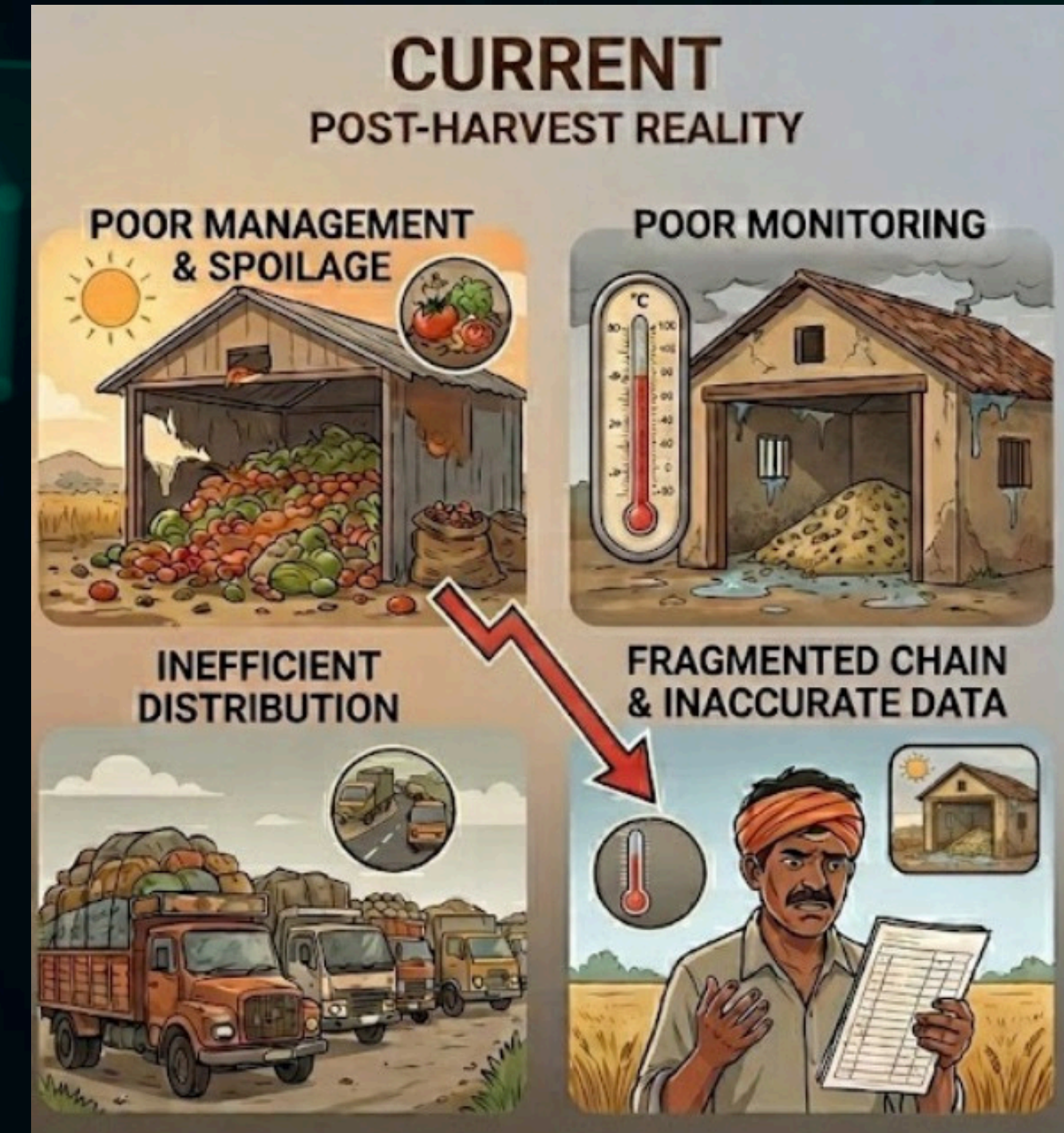
- Python/C++ (Backend), React (Frontend), IoT Sensors, Cloud Database.
- Temperature & humidity sensors installed in warehouses for continuous monitoring.
- Live dashboard for stock levels, storage conditions & instant alerts.
- Data Collection → AI Analysis → Alert Generation → Smart Storage/Distribution Suggestions.
- Data Input → Environment Monitoring → Dashboard Updates → Loss Reduction via Smart Planning.



IMPACT AND BENEFITS



- Reduces post harvest losses by preventing spoilage through real time monitoring and early alerts.
- Increases farmers income as more produce safely reaches the market
- Improves food availability by reducing waste in the supply chain.
- Enhances supply chain efficiency with smart storage and distribution planning.
- Promotes sustainable agriculture by minimizing food waste and better resource usage.





Testing & Validation

- Studies show that a large percentage of agricultural produce in India is lost due to poor storage and transportation facilities.
- Government reports highlight the need for better cold storage infrastructure and supply chain modernization.
- Research papers emphasize the role of IoT and AI in reducing post harvest losses.
- Case studies demonstrate that real time monitoring systems significantly reduce spoilage in warehouses.
- Sustainable development reports recommend smart inventory and distribution planning to improve food security.



RESEARCH AND REFERENCES



- https://ishs.org/ishs-article/712_100/1 — Postharvest losses due to gaps in cold chain in India — quantifies losses and cold-chain infrastructure gaps.
- <https://epubs.icar.org.in/index.php/AERR/article/view/1409862> — Assessment of harvest and post-harvest losses of major pulses in India — national survey quantifying storage and post-harvest losses.
- Postharvest losses due to gaps in cold chain in India – ISHS https://ishs.org/ishs-article/712_100/
- An IoT-Based Controlled Environment Storage to Prevent Onion Post-Harvest Spoilage (arXiv) <https://arxiv.org/abs/2601.10745>
- <https://www.journaljeai.com/index.php/JEAI/article/view/38133> — Post-Harvest Losses in Mango Supply Chains in India — review covering loss by stage (harvest, transit, storage).



THANK YOU